# Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

**Ringkasan**

PENDAHULUAN
TINJAUAN PUSTAKA
[11pt,a4paper,twoside]article
fontspec Verdana
graphicx ulem amsmath tikz forest comment lscape rotating lastpage fancyhdr ifthen changepage algorithmic algorithm enumerate
shapes.multipart,calc decorations.text,calc,arrows.meta chains,fit,shapes automata,positioning
1 COMPXXXX-E1
>1 *Turn over* COMPXXXX-E1

## The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL X MODULE, AUTUMN/SPRING SEMESTER 20XX-20XX

**«MODULE NAME»**

Time allowed: X Hours and Y Minutes

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*Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced*

***«Answer Rubric»***

*«Calculator Rubric»*

*Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.*

*No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.*

***DO NOT turn examination paper over until instructed to do so***

**ADDITIONAL MATERIAL:** «List Additional/Supplementary Materials MUST be specified in this section of the paper Handouts, MCQ, number of answer books x , graph paper, statistical tables (Neaves, SI, E&E), statutes, Books allowed, None».
**INFORMATION FOR INVIGILATORS:** «to include any announcements or to indicate if the paper can be taken away at the end of the examination / None».

SECTION A

red**EXAMPLE 1**
**1. Topic 1: «Type_Topic».0**

[a. ]<Type_Text>
1. [< < > > Marks]

2. <Type_Text>

[< < > > Marks]

3. <Type_Text>

[< < > > Marks]

**End of Question 1: Total < < > > marks**

red**EXAMPLE 2**
**Question 1:** <Type_Topic>**.** <Type_Text>

[**overall «» marks**]

[a. ]<Type_Text>
1. [< < > > Marks]

2. <Type_Text>

[< < > > Marks]

3. <Type_Text>

[< < > > Marks]

**Question 2:** <Type_Topic>**.** <Type_Text>

[**overall < < > > marks**]

**Either** (a) <Type_Text>

[< < > > Marks]

**Or** (b) <Type_Text>

[< < > > Marks]

**Question 3:** <Type_Topic>**.** <Type_Text>

[**overall < < > > marks**]

**Question 4:** <Type_Topic>**.** <Type_Text>

[**overall < < > > marks**]

SECTION B

**Question 5:** <Type_Topic>**.** Answer **two** parts of this question, each carry < < > > marks.

[**overall < < > > marks**]

(a) <Type_Text>
(b) <Type_Text>
(c) <Type_Text>
(d) <Type_Text>
(e) <Type_Text>
(f) <Type_Text>
(g) <Type_Text>
(h) <Type_Text>

**Question 6:** <Type_Topic>**.** <Type_Text>

[1. ]<Type_Text>

1. [< < > > Marks]

2. <Type_Text>

[< < > > Marks]

3. <Type_Text>

[< < > > Marks]

[**overall** < < > > **marks**]

**Question 7:** <Type_Topic>**.** <Type_Text>

[**overall** < < > > **marks**]

**Question 8:** <Type_Topic>**.** <Type_Text>

[**overall** < < > > **marks**]

**Question 9:** <Type_Topic>**.** <Type_Text>

[**overall** < < > > **marks**]

**Question 10:** <Type_Topic>**.** <Type_Text>

[**overall** < < > > **marks**]

*The following pages contain more examples about the formatting and provide a domain specific template for computer science.*

SECTION A

**Question 1:**

**[overall XX marks]**

Answer to the following question with graphics

[shorten >=2pt,node distance=3cm,on grid,auto] [state,initial] (Q1) $Q_1$; [state] (Q2) [right=of Q1] $Q_2$; [state] (Q3) [right=of Q2] $Q_3$; [state] (Q4) [right =of Q3] $Q_4$; [state] (Q5) [below =of Q1] $Q_5$; [state, accepting] (Q6) [right =of Q5] *Accept*; `->](Q2)edge`loop above] node $\begin{array}{l} a \to a, R \\ y \to y, R \end{array}$ () (Q3) edge [loop above] node $\begin{array}{l} b \to b, R \\ z \to z, R \end{array}$ () (Q4) edge [loop above] node $\begin{array}{l} b \to b, L \\ z \to z, L \\ a \to a, L \\ y \to y, L \end{array}$ () (Q5) edge [loop below] node $\begin{array}{l} y \to y, R \\ z \to z, R \end{array}$ () (Q1) edge node $a \to x, R$ (Q2) (Q2) edge node $b \to y, R$ (Q3) (Q3) edge node $c \to z, L$ (Q4) (Q4) edge [bend left] node $x \to x, R$ (Q1) (Q1) edge [left] node $y \to y, R$ (Q5) (Q5) edge [above] node $\sqcup \to \sqcup, R$ (Q6);

**Question 2:**

**[overall XX marks]**

Answer the following question involving the following graphics and divided into sub-questions.

(O) at (0,0); [xshift=6cm] (O) at (0,0); [fill=gray!30] (O) circle (3.9); [fill=red!30] (O) circle (3.1); [fill=green!40] (O) circle (2.3); [fill=yellow!70] (O) circle (1.5); [fill=blue!45] (O) circle (0.7); [decoration=text along path,reverse path,text align=align=center,text=FSM,decorate] (0.9,0.1) arc (0:180:0.9); [decoration=text along path,reverse path,text align=align=center,text=Pushdown Automata,decorate] (1.6,0.2) arc (0:180:1.6); [decoration=text along path,reverse path,text align=align=center,text=Turing Machines,decorate] (2.1,0.5) arc (0:180:2.1); [decoration=text along path,reverse path,text align=align=center,text=Undecidable,decorate] (2.9,0.5) arc (0:180:2.9);

1. this is one question with inline equations $\sum_{j=1}^{n} x_j \log(x_j)$

[X Marks]

2. this is one question with equations $\mathrm{y} \Longleftrightarrow \mathbf{Ax} \equiv \{ y_1 = a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n$
   $a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n$
   $\ldots$
   $a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n$

[X Marks]

3. this is a question about a Table

$\delta$:

| | 0 | 1 |
|---|---|---|
| $A$ | $A$ | $A, B$ |
| $B$ | $C$ | $C$ |
| $C$ | $\emptyset$ | $\emptyset$. |

[X Marks]

SECTION B

**Question 3:**

**[overall 20 marks]**

This is a question about pseudocodes

General Differential Evolution Framework [1] Generate an initial population of $Np$ individuals Evaluate fitness of each solution in population $Np$ termination condition is not met each $\mathbf{x^i}$ in $Np$ Generate provisional offspring $\mathbf{x^{off'}}$ by mutation Generate offspring $\mathbf{x^{off}}$ by crossover Evaluate fitness of $\mathbf{x^{off}}$ Make a note whether $\mathbf{x^i}$ or $\mathbf{x^{off}}$ has a better performance each $\mathbf{x^i}$ in $Np$ Perform all the replacements by choosing the best between parent offspring

IMPORTANT NOTE:

COMPILE WITH

***XeLaTex***

AND NOT

***LaTex*** OR ***pdfLaTex***